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Equivalence

Def 2:
If $\Delta(m,n) = \Delta(m',n')$, we say (m,n) & (m',n') are net difference equivalent.
In this case we write $(m,n) \sim (m',n')$

idea:

If I have \$50 & owe \$100, my net dif $= -50$

$$\Delta(50,100) = (0,50)$$

If I have \$1 & owe \$51, net dif $= -50$

$$\Delta(1,51) = (0,50)$$

$\Rightarrow$ These are the same if we're talking about how much I owe.

To formalize this, I define the equivalence class:

$$[0,50] = \{(m,n) \in \mathbb{N} : \Delta(m,n) = (0,50)\}$$

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Generally define

$$[m,n] = \{(a,b) \in \mathbb{N} \times \mathbb{N} : \Delta(a,b) = \Delta(m,n)\}$$

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Generally define

$$[m,n] = \{(a,b) \in \mathbb{N}\times\mathbb{N} : \Delta(a,b) = \Delta(m,n)\}$$

representatives

$$[0,50] = [30,80],\ [5,55],\ [10,6]$$

different reps., same result

equality of sets

$$\Delta(0,50) = (0,50) \neq (5,55)$$

$$\hookrightarrow = \Delta(5,55)$$

must take Δ of both

$(m,n) \sim (a,b)$ if $\Delta(m,n) = \Delta(a,b)$

$$[30,80] = \{(a,b) \in \mathbb{N}\times\mathbb{N} : (a,b) \sim (30,80)\}$$

The Integers are:

$$\mathbb{Z} = \{a : a = [m,n] \text{ for some } (m,n) \in \mathbb{N}\times\mathbb{N}\}$$

- basically the set of all equivalence classes

Def. $[m,n] + [r,s] = [m+r, n+s]$

If $[m,n] = [m',n']$ & $[r,s] = [r',s']$,

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The Integers are:

$$\mathbb{Z} = \{ a : a = [m,n] \text{ for some } (m,n) \in \mathbb{N} \times \mathbb{N} \}$$

- basically the set of all equivalence classes

Def. $[m,n] + [r,s] = [m+r, n+s]$

If $[m,n] = [m',n']$ & $[r,s] = [r',s']$,

then $[m,n] + [r,s] = [m'+r', n'+s']$

<u>ex</u>

$[100,0] = [200,100]$

$[0,12] = [32,44]$ ← this is $\cdot 12$

$[100,0] + [0,12] = [200,100] + [32,44]$

$[100,12] \overset{?}{=} [232,144]$

$100-12 \qquad 232-144$

$(88,0) = (88,0)$

$\boxed{\Delta[100,12] = \Delta[232,144] = (88,0).}$